Daily Logic Drill #5

Sheet by David Akinyele-Aje

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Spot-the-Flaw · Proof Critique · Invalid Algebraic Operations · Fallacious Logic Steps.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition (≤15 s each)

Identify the exact logical or algebraic fallacy. Pure recognition.

1. In the proof step "If $a = b$, then $a(a-b) = (a-b)^2 \implies a = a - b$", what operation was invalidly performed?

2. True or false: "$\sqrt{x^2} = x$ for all real numbers x."

3. What fallacy is committed when one assumes the conclusion P in order to prove P?

4. A student writes: "$x^2 > 9 \implies x > 3$." What case did the student miss?

5. True or false: "If $ac \equiv bc \pmod{m}$, then $a \equiv b \pmod{m}$ for all non-zero integers c, m."

6. In the deduction "If $x^2 = 4$, then $x = 2$", which direction of logical implication was omitted?

7. Identify the error in: "Since $-1 = (-1)^1 = (-1)^{2/2} = ((-1)^2)^{1/2} = 1^{1/2} = 1$."

8. True or false: "Proving that the converse of a statement is true proves the original statement."

9. What is invalid about dividing both sides of $x^2 - x = 0$ by x to conclude $x = 1$?

10. A student claims: "$0 \times 1 = 0 \times 2$, so cancelling 0 gives $1 = 2$." Which field axiom is violated?

Section B Manipulation Drills (≤50 s each)

Locate the exact line number ($1, 2, 3, \ldots$) where the proof breaks down.

1. Consider the following attempt to prove $2 = 1$:
Line 1: Let $a = b$.
Line 2: Then $a^2 = ab$.
Line 3: Subtract b^2: $a^2 - b^2 = ab - b^2$.
Line 4: Factorise: $(a-b)(a+b) = b(a-b)$.
Line 5: Divide by $(a-b)$: $a + b = b$.
Line 6: Substitute $a = b$: $b + b = b \implies 2b = b$.

Line 7: Divide by b: $2 = 1$.
Which line contains the first invalid step?

2. Consider the attempt to prove $1 = -1$:
Line 1: $-1 = \frac{-1}{1}$.
Line 2: Taking square roots: $\sqrt{-1} = \sqrt{\frac{-1}{1}}$.
Line 3: $\sqrt{-1} = \frac{\sqrt{-1}}{\sqrt{1}}$.
Line 4: Also $\sqrt{-1} = \sqrt{\frac{1}{-1}} = \frac{\sqrt{1}}{\sqrt{-1}}$.
Line 5: Equating gives $\frac{\sqrt{-1}}{1} = \frac{1}{\sqrt{-1}}$.
Line 6: Cross-multiplying gives $(\sqrt{-1})^2 = 1 \times 1 \implies -1 = 1$.
Which line contains the first invalid manipulation of square roots?

3. Consider the argument:
Line 1: We want to show $x^2 + 1 > 0$ for all real x.
Line 2: Assume $x^2 + 1 > 0$.
Line 3: Subtract 1: $x^2 > -1$.
Line 4: Since $x^2 \geq 0$ for all real x, $x^2 > -1$ is true.
Line 5: Therefore $x^2 + 1 > 0$ is proved.
What is the logical defect in this structure?

4. Consider the proof attempt for: "For all real x, if $x^2 > 4$, then $x > 2$."
Line 1: Suppose $x^2 > 4$.
Line 2: Take square roots: $\sqrt{x^2} > \sqrt{4}$.
Line 3: Simplify: $x > 2$.
Line 4: Hence the claim is proved.
Which line contains the invalid simplification, and what should it be?

5. Consider the argument:
Line 1: Let $S = 1 + 2 + 4 + 8 + 16 + \dots$
Line 2: Then $2S = 2 + 4 + 8 + 16 + \dots$
Line 3: Subtracting gives $S - 2S = 1 \implies -S = 1 \implies S = -1$.
Line 4: Therefore $1 + 2 + 4 + 8 + \cdots = -1$.
Which line relies on an invalid assumption about infinite series?

6. Consider the attempt to prove every integer is even:
Line 1: Let $P(n)$ be the statement "n is even".
Line 2: Assume $P(k)$ is true, so k is even, i.e., $k = 2m$.
Line 3: Then $k + 2 = 2m + 2 = 2(m + 1)$, which is even, so $P(k + 2)$ is true.
Line 4: Since $P(0)$ is true ($0 = 2 \times 0$), $P(n)$ holds for all even n.
Line 5: Therefore all integers are even.
Which line makes an unjustified logical leap?

7. Consider:
Line 1: $a < b$.
Line 2: Multiply both sides by c: $ac < bc$.
Line 3: Subtract ac: $0 < bc - ac = c(b - a)$.
Line 4: Divide by $b - a$ (which is > 0): $0 < c \implies c > 0$.
What condition on c was implicitly assumed in Line 2?

8. Consider the argument:
Line 1: To prove $a = b$, it suffices to show $a^2 = b^2$.
Line 2: Take $a = 3$ and $b = -3$.
Line 3: $3^2 = (-3)^2 = 9$.
Line 4: Therefore $3 = -3$.
Which line contains a false logical equivalence?

9. Consider the proof attempt for $\sqrt{2} + \sqrt{3} < \sqrt{10}$:
Line 1: Square both sides: $(\sqrt{2} + \sqrt{3})^2 < 10$.
Line 2: Expand: $2 + 2\sqrt{6} + 3 < 10 \implies 5 + 2\sqrt{6} < 10$.
Line 3: Subtract 5: $2\sqrt{6} < 5$.
Line 4: Square both sides: $4 \times 6 < 25 \implies 24 < 25$.
Line 5: Since $24 < 25$ is true, the original inequality holds.
Why is this argument incomplete as written?

10. Consider the argument:
Line 1: $1^2 = 1$.
Line 2: $2^2 = 1 + 3$.
Line 3: $3^2 = 1 + 3 + 5$.
Line 4: By pattern recognition, n^2 is the sum of the first n odd positive integers.
Line 5: Therefore the theorem is proved for all $n \in \mathbb{Z}^+$.
Why is pattern recognition up to $n = 3$ not a rigorous proof?

Section C Substitution & Structure ($\leq$90 s each)

Analyze proof critiques with tricky multiple-choice options.

1. A student attempts to prove: "For all real x, if $x^3 > x$, then $x > 1$." *(after TMUA 2021 Paper 2 Q5)*
Proof Attempt:
Line 1: Assume $x^3 > x$.
Line 2: Factorise: $x(x^2 - 1) > 0$.
Line 3: Divide by x: $x^2 - 1 > 0$.
Line 4: Therefore $x^2 > 1$, which implies $x > 1$.
Which of the following best describes the errors in this proof?

A) Line 3 is invalid when $x < 0$, and Line 4 misses the possibility $x < -1$.

B) Line 2 is incorrect factorisation.

C) Line 3 is invalid for all x, but Line 4 is correct.

D) The proof is completely correct.

E) Line 1 is an invalid starting point.

2. Consider the statement: "If n is an integer such that n^2 is divisible by 6, then n is divisible by 6." *(after TMUA 2018 Paper 2 Q6)*
Attempted Proof:
Line 1: Let $n^2 = 6k$ for some integer k.
Line 2: $n = \sqrt{6k} = \sqrt{6}\sqrt{k}$.
Line 3: Since n is an integer, $\sqrt{6k}$ must be an integer, so $k = 6m^2$ for some integer m.
Line 4: Then $n = \sqrt{6 \times 6m^2} = 6m$, so n is divisible by 6.
Is this proof valid?

A) Yes, the proof is fully correct and rigorous.

B) No, Line 3 assumes what it tries to prove by asserting $k = 6m^2$.

C) No, Line 2 is invalid because square roots cannot be applied to integers.

D) No, Line 4 contains an arithmetic error in $\sqrt{36m^2}$.

E) No, the original statement itself is false.

3. A student writes a proof of the claim: "For all positive integers n, $n^2 + 5n + 6$ is even."
Line 1: $n^2 + 5n + 6 = (n + 2)(n + 3)$.
Line 2: One of $n + 2$ or $n + 3$ is even, and the other is odd.
Line 3: The product of an even integer and an odd integer is always even.
Line 4: Therefore $(n + 2)(n + 3)$ is even for all positive integers n.
Which evaluation of this proof is correct?

A) The proof is completely valid and correct.

B) Line 1 is an incorrect factorisation.

C) Line 2 is false because $n + 2$ and $n + 3$ can both be even.

D) Line 3 is false because even $\times$ odd is odd.

E) Line 4 does not follow because n must be specified as even or odd first.

4. A student attempts to prove: "For all real x, $x^2 + 4x + 5 > 0$."
Line 1: Completing the square: $x^2 + 4x + 5 = (x + 2)^2 + 1$.
Line 2: Since $(x + 2)^2 \geq 0$ for all real x, $(x + 2)^2 + 1 \geq 1 > 0$.
Line 3: Thus $x^2 + 4x + 5 > 0$ for all real x.
Which evaluation of this proof is correct?

A) The proof is completely valid and correct.

B) Line 1 is incorrect because $(x + 2)^2 + 1 = x^2 + 4x + 3$.

C) Line 2 is invalid because $(x + 2)^2$ can be negative for $x < -2$.

D) Line 3 is invalid because ≥ 1 does not imply > 0.

E) The statement itself is false for $x = -2$.

5. A student attempts to prove: "If $a > b > 0$, then $a^2 > b^2$."
Line 1: Start with $a > b$.
Line 2: Since $a > 0$ and $b > 0$, multiply Line 1 by a: $a^2 > ab$.
Line 3: Multiply Line 1 by b: $ab > b^2$.
Line 4: By transitivity, $a^2 > ab > b^2 \implies a^2 > b^2$.
Which evaluation of this proof is correct?

A) The proof is completely valid and correct.

B) Line 2 is invalid because multiplying by a reverses the inequality.

C) Line 3 is invalid because b could be negative.

D) Line 4 is invalid because transitivity does not apply to $>$.

E) The statement is false for $a = 2, b = 1$.

6. Consider the attempt to prove: "If x^2 is irrational, then x is irrational."
Line 1: We prove the contrapositive: "If x is rational, then x^2 is rational."
Line 2: Let $x = p/q$ where $p, q \in \mathbb{Z}$ and $q \neq 0$.
Line 3: Then $x^2 = p^2/q^2$.
Line 4: Since $p, q \in \mathbb{Z}$, $p^2, q^2 \in \mathbb{Z}$ and $q^2 \neq 0$, so $x^2 \in \mathbb{Q}$.
Line 5: Since the contrapositive is true, the original statement is true.
Which evaluation of this proof is correct?

A) The proof is completely valid and correct.

B) Line 1 is incorrect because the contrapositive of $P \implies Q$ is $Q \implies P$.

C) Line 3 is incorrect algebra.

D) Line 4 is invalid because p^2/q^2 might not be in lowest terms.

E) The original statement is false.

7. A student attempts to prove: "For all real $x \neq 1$, $\frac{x^2-1}{x-1} = x + 1$."
Line 1: $\frac{x^2-1}{x-1} = \frac{(x-1)(x+1)}{x-1}$.
Line 2: Cancel $(x - 1)$ from numerator and denominator: $= x + 1$.
Line 3: Therefore the identity holds for all $x \neq 1$.
Which evaluation of this proof is correct?

A) The proof is completely valid and correct.

B) Line 2 is invalid because cancellation is never allowed in rational expressions.

C) Line 3 is invalid because $x = 1$ was excluded.

D) Line 1 is an incorrect factorisation.

E) The claim is false for $x = 2$.

8. Consider the claim: "For all integers n, $n^3 - n$ is divisible by 3."
Proof Attempt:
Line 1: $n^3 - n = n(n^2 - 1) = (n - 1)n(n + 1)$.
Line 2: $(n - 1), n, (n + 1)$ are three consecutive integers.
Line 3: In any three consecutive integers, exactly one is a multiple of 3.
Line 4: Therefore their product is divisible by 3.
Which evaluation of this proof is correct?

A) The proof is completely valid and correct.

B) Line 1 is an incorrect factorisation.

C) Line 2 is false for negative integers n.

D) Line 3 is false because two consecutive integers can be multiples of 3.

E) Line 4 does not follow because divisibility by 3 requires $n > 0$.

Section D Challenge (TMUA / SMC difficulty)

Subtle proof critiques from authentic competition sources.

1. Consider the following attempt to prove: "For all real numbers x, if $x^2 - 5x + 6 = 0$, then $x = 2$."
(after TMUA 2016 Paper 2 Q11)
Line 1: $x^2 - 5x + 6 = 0 \implies (x - 2)(x - 3) = 0$.

Line 2: This means $x = 2$ or $x = 3$.
Line 3: Substituting $x = 2$: $2^2 - 5(2) + 6 = 4 - 10 + 6 = 0$, which is true.
Line 4: Therefore $x = 2$.
Which statement best describes this proof attempt?

A) The proof is correct.

B) Line 2 does not follow from Line 1.

C) Line 4 does not follow because $x = 3$ is also a solution, so $x^2 - 5x + 6 = 0$ does not imply $x = 2$ exclusively.

D) Line 3 contains an arithmetic error.

E) Line 1 is an incorrect factorisation.

2. A student attempts to prove: "All horses are the same color" using induction.
Base Case: In any set of 1 horse, all horses in the set have the same color. (True)
Inductive Step: Assume that in any set of k horses, all horses have the same color. Now consider a set of $k + 1$ horses $\{h_1, h_2, \ldots, h_k, h_{k+1}\}$.
- The subset $\{h_1, h_2, \ldots, h_k\}$ has k horses, so by IH they all have the same color.
- The subset $\{h_2, h_3, \ldots, h_{k+1}\}$ has k horses, so by IH they all have the same color.
- Since h_2 is in both subsets, all $k + 1$ horses must have the same color.

Where does this famous inductive proof break down?

A) The base case $k = 1$ is false.

B) The transition from $k = 1$ to $k = 2$ fails because the two subsets $\{h_1\}$ and $\{h_2\}$ do not overlap.

C) Induction cannot be applied to finite sets.

D) The inductive hypothesis cannot be assumed for k.

E) The set $\{h_2, \ldots, h_{k+1}\}$ has size $k - 1$, not k.

3. Consider the attempt to prove: "If n^2 is divisible by 3, then n is divisible by 3." *(after TMUA 2017 Paper 2 Q11)*
Line 1: We prove the contrapositive: "If n is not divisible by 3, then n^2 is not divisible by 3."
Line 2: If n is not divisible by 3, then $n = 3k + 1$ or $n = 3k + 2$ for some integer k.
Line 3: If $n = 3k + 1$, $n^2 = 9k^2 + 6k + 1 = 3(3k^2 + 2k) + 1$, which is not divisible by 3.
Line 4: If $n = 3k + 2$, $n^2 = 9k^2 + 12k + 4 = 3(3k^2 + 4k + 1) + 1$, which is not divisible by 3.
Line 5: In both cases n^2 is not divisible by 3, so the contrapositive is true, and the original statement is proved.
Which evaluation of this proof is correct?

A) The proof is completely valid and rigorous.

B) Line 1 states the converse, not the contrapositive.

C) Line 2 misses the case $n = 3k$.

D) Line 4 has incorrect algebra: $(3k + 2)^2 = 9k^2 + 6k + 4$.

E) The original statement is false.

4. A student writes the following proof that $1 = 0$:
Line 1: Let $x = 1$.

Line 2: $x^2 = x$.
Line 3: $x^2 - 1 = x - 1$.
Line 4: $(x-1)(x+1) = x - 1$.
Line 5: Divide by $x - 1$: $x + 1 = 1$.
Line 6: Since $x = 1$, $1 + 1 = 1 \implies 2 = 1$.
Line 7: Subtract 1: $1 = 0$.
Which line contains the fundamental error, and why?

A) Line 3: subtracting 1 is invalid.

B) Line 4: incorrect factorisation.

C) Line 5: division by $x - 1$ is division by zero since $x = 1$.

D) Line 6: substitution of $x = 1$ is invalid.

E) Line 7: subtraction of 1 is invalid.

5. Consider the statement: "For all real numbers x, y, if $x + y > 0$, then $x > 0$ or $y > 0$." *(after TMUA 2020 Paper 2 Q4)*
Attempted Proof by Contrapositive:
Line 1: The contrapositive is: "If $x \leq 0$ and $y \leq 0$, then $x + y \leq 0$."
Line 2: Assume $x \leq 0$ and $y \leq 0$.
Line 3: Adding the two inequalities gives $x + y \leq 0 + 0 = 0$.
Line 4: Since the contrapositive is true, the original statement is proved.
Which evaluation of this proof is correct?

A) The proof is completely valid and correct.

B) Line 1 has the wrong contrapositive because "or" should remain "or".

C) Line 3 is invalid because inequalities cannot be added.

D) Line 4 is invalid because contrapositive proofs are not accepted in mathematics.

E) The original statement is false for $x = 3, y = -1$.

"Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised."

Found a mistake or want to contribute a sheet?
Visit the open-source project at github.com/The-CerealDev/speed-maths